

Output for graph a:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
>  ▾  TERMINAL
● PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI> python -u "c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab4\Graph[a].py"
Node --> Connected Nodes
-----
A-->B, C
B-->A, D, E
C-->A, F
D-->B, G
E-->B, H
F-->C
G-->D
H-->E
❖ PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>
```

Output for graph B:

```
python -u
Node -> (Neighbor: Weight)
-----
S -> (A: 1), (G: 12)
A -> (B: 3), (C: 1)
B -> (D: 3)
C -> (D: 1), (G: 2)
D -> (G: 3)
G -> None (End Node)
❖ PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>
```

Output for graph 3:

```
python -u
Node -> (Neighbor: Weight)
-----
A -> (B: 22), (D: 30), (E: 25)
B -> (A: 22), (C: 20), (D: 11)
C -> (B: 20)
D -> (A: 30), (B: 11), (F: 10)
E -> (A: 25), (F: 25), (G: 40)
F -> (D: 10), (E: 25), (G: 12)
G -> (E: 40), (F: 12), (H: 6)
H -> (G: 6)
❖ PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>
```

Output for the Greedy Best First search :

```
python -u "c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab5>  
--- Greedy Best-First Search ---  
starting point :A  
ending or goal node:H  
Path: A -> B -> D -> G -> H  
saugat tamang  
❖ PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab5>
```

Output for the A star Search Algorithm:

```
python -u "c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab5\INFORMED\a_star.py"

--- A* Search ---
starting point :A
ending or goal node:H
Path: A -> B -> D -> G -> H
Total Actual Cost: 49
saugat tamang
PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab5>
```

Output for the bfs and dfs of graph a {}:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

>  ✓ TERMINAL
  PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>

  • python -u "c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab5\UNINFORMED\bfs_and_dfs.py"

  --- Question (i) ---
  BFS Order for Graph A: ['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H']
  DFS Order for Graph A: ['A', 'B', 'D', 'G', 'E', 'H', 'C', 'F']
  PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>
```

Output for the ucs for graph b (ii):

```
python -u "c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab5\UNINFORMED\ucs.py"

--- Question (ii) ---
UCS Optimal Path to 'G': ['A', 'C', 'G'] with Total Cost: 3
❖ PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>
```

Output for the dls for graph a(iii)

```
python -u "c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab5\UNINFORMED\dls.py"

--- Question (iii) ---
Goal 'F' found within depth limit 2. Path: ['A', 'C', 'F']
❖ PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>
```

Output for simple reflex agent:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
>  TERMINAL
PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI> python -u "c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab6\lab6-1.py"
c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab6\lab6-1.py:45: SyntaxWarning: invalid escape sequence '\['
  print("\[Thorough Check Complete]: Both blocks A and B have been verified as CLEAN!")
Vacuum Agent Simulation(simple reflex agent)
Grid Layout: [A][B]

--- Current Location: Block A ---
What is the condition on block A (clean/dirty) [or 'exit' to quit]: dirty

[Agent Decision]:
  Block A is DIRTY -> Action: Suck

--- Current Location: Block A ---
What is the condition on block A (clean/dirty) [or 'exit' to quit]: clean

[Agent Decision]:
  Block A is CLEAN -> Action: Move_Right

--- Current Location: Block B ---
What is the condition on block B (clean/dirty) [or 'exit' to quit]: dirty

[Agent Decision]:
  Block B is DIRTY -> Action: Suck

--- Current Location: Block B ---
What is the condition on block B (clean/dirty) [or 'exit' to quit]: clean

[Agent Decision]:
  Block B is CLEAN -> Action: Move_Left

--- Current Location: Block A ---
What is the condition on block A (clean/dirty) [or 'exit' to quit]: clean

[Agent Decision]:
  Block A is CLEAN -> Action: Move_Right
\[Thorough Check Complete]: Both blocks A and B have been verified as CLEAN!
Ending simulation
PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>
```

Output for the n-queen problem :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
>  TERMINAL
PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI> python -u "c:\Saugat-Tamang-024BSCIT036\4th sem\AI\lab6\lab6-2.py"
--- N-Queens Solver (8x8 Board) ---

Found Solution:
[Q] [.] [.] [.] [.] [.] [.] [.]
[.] [.] [.] [.] [.] [.] [Q] [.]
[.] [.] [.] [.] [Q] [.] [.] [.]
[.] [.] [.] [.] [.] [.] [.] [Q]
[.] [Q] [.] [.] [.] [.] [.] [.]
[.] [.] [.] [Q] [.] [.] [.] [.]
[.] [.] [.] [.] [.] [Q] [.] [.]
[.] [.] [Q] [.] [.] [.] [.] [.]
PS C:\Saugat-Tamang-024BSCIT036\4th sem\AI>
```